

# **The Bias-Variance Tradeoff and Evaluation Metrics**

Prof. Artur Jordão

# Learning and Testing Phase

# Definitions

## Learning and Testing Phase

- Define  $D = \{(x_i, y_i)\}_{i=1}^n$  a dataset
- Let  $D_{train}$  be a subset of  $D$  such that  $D_{train} \subseteq D$ 
  - Samples and their (seen) labels
- Let  $D_{test}$  be a subset of  $D$  such that  $D_{test} \subseteq D$ 
  - In practice, unseen labels
- Important properties
  - $D = D_{train} \cup D_{test}$
  - $D_{train} \cap D_{test} = \emptyset$
  - Data leak  $D_{train} \cap D_{test} \neq \emptyset$

# Overview

## Learning and Testing Phase

### Learning Phase

(Here  $X$  and  $Y$  come from  $D_{train}$ )

```
model.fit(X, Y)
```

$$(X^T X)^{-1} X^T Y$$

$$\mathbf{w}_i \leftarrow \mathbf{w}_i - \eta \frac{\partial}{\partial \mathbf{w}_i} \mathcal{L}(\mathbf{W})$$

$$\mathbf{w}_i \leftarrow \mathbf{w}_i - \eta \frac{1}{|\beta|} \sum_{j=\beta_t}^n \frac{\partial}{\partial \mathbf{w}_i} \mathcal{L}^j(\mathbf{W})$$

### Testing Phase

(Here  $X$  comes from  $D_{test}$ )

```
y_pred = model.predict(X)
```

$$\hat{Y} = XW^T + b$$

# Quality of the Learning Trajectory

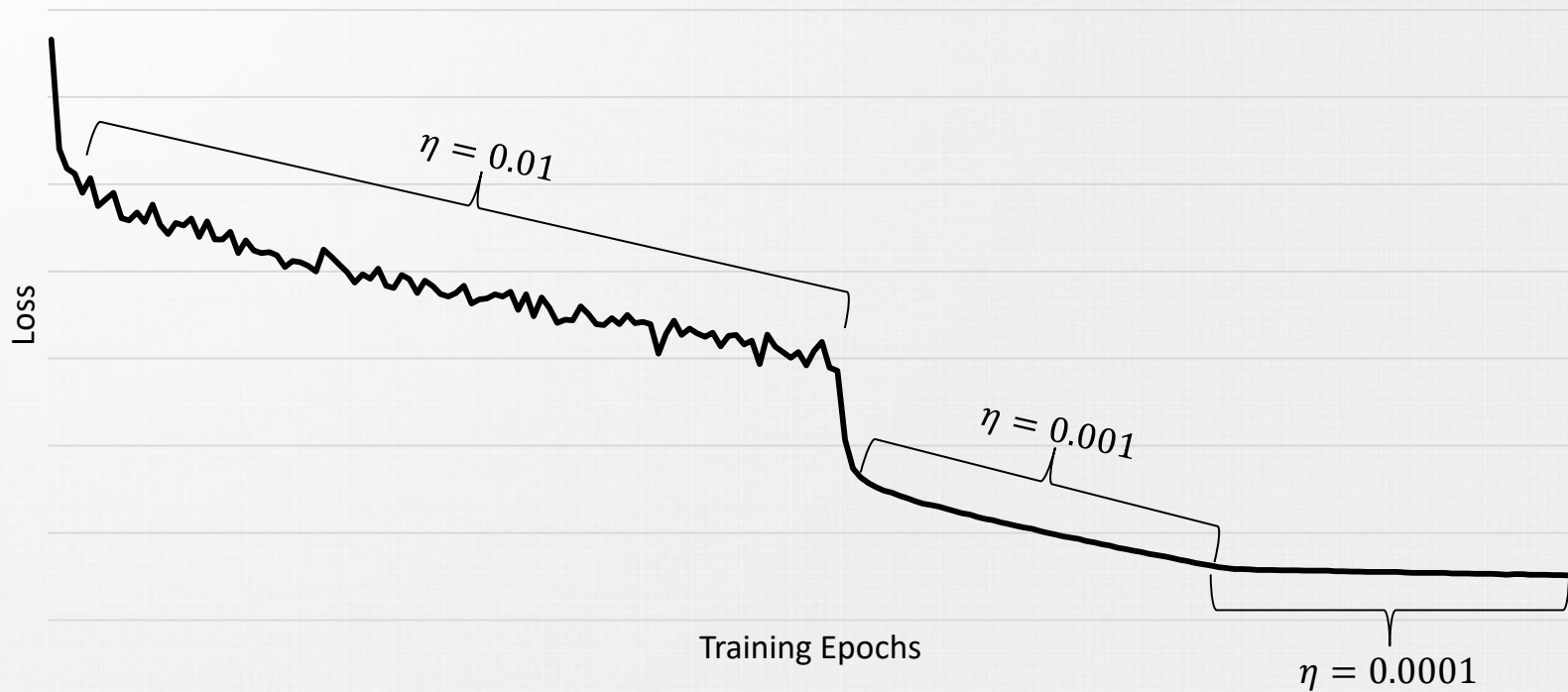
## Learning and Testing Phase

- How to measure the quality of the training trajectory?
  - Dynamics of training
- Loss curve
- Decision boundary
- Loss landscape (Li et al., 2018)

# Loss Curve

## Learning and Testing Phase

- Early stop



# Decision Boundary

## Learning and Testing Phase

- Decision boundary (a.k.a. decision surface) is a boundary (line or surface, in higher dimensions) that separates the examples of different classes

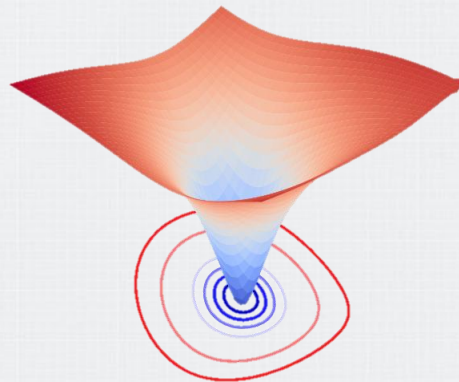
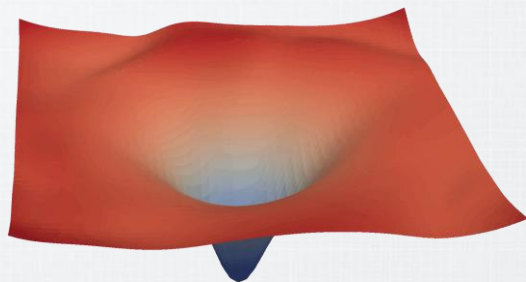
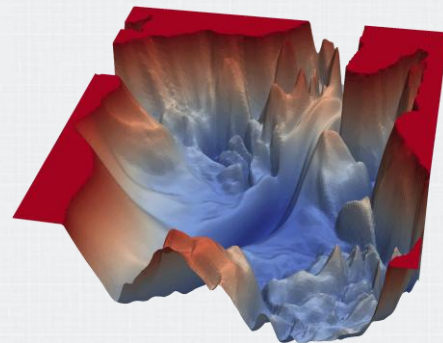
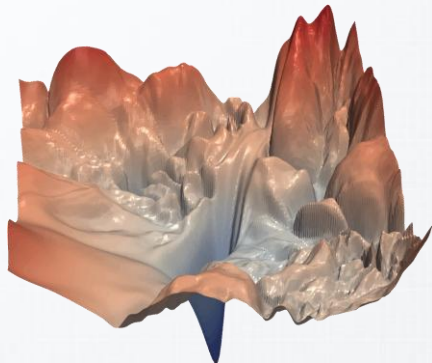
[Hands-on](#)



# Loss Landscape

## Learning and Testing Phase

- <http://www.telesens.co/loss-landscape-viz/viewer.html>



# Testing Phase

## Learning and Testing Phase

- Once the training is done, how can we measure the predictive ability of the model?
- Quantitative metrics for predictive ability
  - Accuracy
  - Precision
  - Recall
  - Confusion Matrix
  - Loss
  - Pearson Correlation
  - Pairwise

# Empirical Risk

## Learning and Testing Phase

- Independent and Identically Distributed (i.i.d) samples
  - Training and testing samples drawn independently from the same distribution
  - We usually develop models under the assumption that the *training and testing data are i.i.d*
- In practical scenarios, the distribution of  $\mathbf{D}_{test}$  is unknown, therefore, we estimate the empirical risk (or training error)
  - The proportion of training samples where predictions do not match the real values
  - The expected (or population) risk is the error on  $\mathbf{D}_{test}$
- Requirements to empirical risk
  - The model  $\mathcal{F}$
  - The data  $X, Y$
  - A loss function  $\mathcal{L}$

# Benchmark

## Learning and Testing Phase

- A benchmark provides predefined training (optional) and testing sets
  - It allows us to evaluate different methods under the **same conditions**
  - Examples: Kaggle, CIFAR-10, GSM8K, HumanEval
- The metric depends on the application
  - **Some benchmarks have their own metrics**
  - For example, accuracy on CIFAR-10 and Top-5 accuracy/error on ImageNet

# Predictive Ability Metrics

## Learning and Testing Phase

- In binary classification, predictions fall into four cases

| True class | Predicted Class         |                         |       |
|------------|-------------------------|-------------------------|-------|
|            | Positive                | Negative                | Total |
| Positive   | true positive ( $tp$ )  | false negative ( $fn$ ) | $p^*$ |
| Negative   | false positive ( $fp$ ) | true negative ( $tn$ )  | $n^*$ |
| Total      | $p'$                    | $n'$                    | $n$   |

# Predictive Ability Metrics

## Learning and Testing Phase

- Common performance measures for two-class problems

| Metric            | Formula                         |
|-------------------|---------------------------------|
| Error             | $(fp + fn)/n$                   |
| Accuracy          | $\frac{tp + tn}{n} = 1 - error$ |
| Tp-rate or Recall | $tp/p^*$                        |
| Fp-rate           | $fp/n^*$                        |
| Precision         | $tp/p'$                         |
| Sensitivity       | Tp-rate                         |
| Specificity       | $1 - Fp-rate$                   |

# Predictive Ability Metrics

## Learning and Testing Phase

- A common performance measure for multiclass problems is the confusion matrix
  - A matrix whose entry  $(i, j)$  contains the **number of samples** that a model assign the class  $i$  to the class  $j$
  - The diagonal indicates correct predictions ( $tp + tn$ )
  - Off-diagonals indicates classification errors ( $fn + fp$ )
- The confusion matrix is very useful because it allows us to understand which classes the model is confusing

|       | $\hat{y}_1$ | $\hat{y}_2$ | $\hat{y}_3$ |
|-------|-------------|-------------|-------------|
| $y_1$ | 9           | 4           | 1           |
| $y_2$ | 4           | 3           | 0           |
| $y_3$ | 1           | 0           | 2           |

[Hands-on](#)



# Predictive Ability Metrics

## Learning and Testing Phase

- Top- $k$  accuracy
  - Suitable when dealing with many classes
  - $\hat{y}$  must give a “confidence” for each possible class
  - Consider the classification as correct if the true label is among the top- $k$  most confident predictions

| $\hat{y}$         | $y$ | Top-1 | Top-2   | Top-3   |
|-------------------|-----|-------|---------|---------|
| [0.2 0.4 0.3 0.1] | 2   | Wrong | Correct | Correct |
| [0.6 0.3 0.1 0.0] | 3   | Wrong | Wrong   | Correct |
| [0.5 0.3 0.1 0.1] | 1   | Wrong | Correct | Correct |

[Hands-on](#)



# Predictive Ability Metrics

## Learning and Testing Phase

- Receiver operating characteristic (ROC) curve
  - It plots curve of **false positives** versus **true positives** for each value of the hyperparameter (i.e., decision threshold)
- The Area Under the Curve (AUC) ROC curve is useful
  - Since it help us to visualize values that would be good choices for the tradeoff

# **Overfitting, Underfitting, Generalization and No Free Lunch**

# Generalization

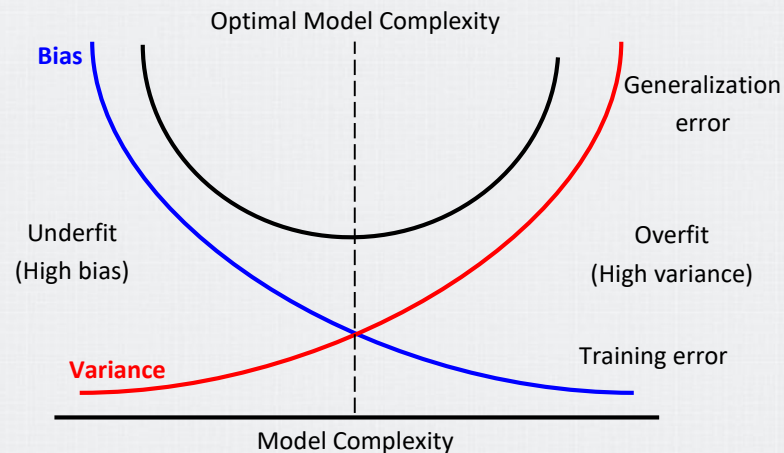
## Overfitting, Underfitting, Generalization and No Free Lunch

- Generalization
  - The quality of the model in predicting new (unseen) data
- Overfitting
  - The model performs well on training but poorly on testing/validation
  - Complex models tend to (we can avoid/handle this) overfit the data
- Underfitting
  - The model fails to find a pattern in the data

# The Bias-Variance Tradeoff

Overfitting, Underfitting, Generalization and No Free Lunch

- Bias
  - Low bias: the model predicts well the samples of the training data
  - High bias: the model makes many mistakes in the training data
- Variance
  - Error of the model due to its sensitivity to small fluctuations in the training set
- Bias–variance tradeoff
  - Low-bias hypotheses that fit the training data well
  - Low-variance hypotheses that **may** generalize better
- U-shaped curve



# No Free Lunch Theorem

## Overfitting, Underfitting, Generalization and No Free Lunch

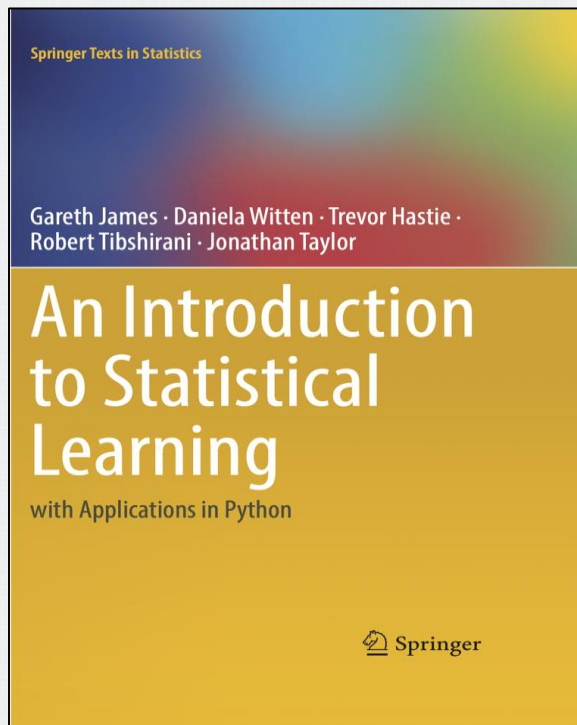
- The *no free lunch theorem* states that every learning algorithm is as good as any other when averaged over all possible problems (Wolpert, 1996)
  - **There is no universal learning algorithm able to solve all tasks precisely**
- Under a uniform distribution over problems (search/learning problems), all algorithms perform equally
  - A particular model or algorithm is **better** than average on some problems, it must be **worse** than average on others

# Bibliography



# Bibliography

- An Introduction to Statistical Learning with Applications in Python
  - Chapter 4
    - 4.4.3 Linear Discriminant Analysis for  $p > 1$



# Bibliography

- INTRODUCTION TO Machine Learning
  - Chapter 19
    - 19 Design and Analysis of Machine Learning Experiments

